Post-Mid-Semester Research Assignment Prompts 01 and 02

AI Tools for Research

Evaluating Tool-Selection Accuracy in Multi-Tool Agentic AI Systems

Submission field	Details
Student	Omkar A. Mohije
Course	AI Tools for Research
Institution	Indian Institute of Information Technology Allahabad
Submission date	22 October 2026
Scope	Completed Prompt 01 and Prompt 02 only
Document type	Audited record of AI-assisted research planning; not an experimental study

Evidence caution: This report distinguishes statements observable in the supplied review and conversation from assertions about primary publications. No implementation, model run, or experimental result is claimed.

1. Assignment Scope and Objectives

The official *AI Tools for Research* post-mid-semester assignment describes a transition from literature review to an eventual implementation-based study. For this submission, the user reports a later instructor announcement restricting the deadline of 22 October 2026 to Prompts 01 and 02. The announcement itself was not independently supplied; its scope and deadline are recorded as user-provided instructions [A1].

Prompt	Official prescribed output	What this report contains
01 - Accountable workspace	Workspace and decision log; review scope	Verbatim submitted prompt; reviewed scope; workflow; D001-D007 and open questions.
02 - Evidence-based research gaps	Gap matrix: source evidence -> candidate question -> possible experiment	Methods/datasets/metrics; five testable candidates; evidence provenance; top-three ranking.

Later prompts (literature-to-experiment selection, baseline validation, coding, experiments, paper writing) remain **outside this submission**. Google Colab (free) is the stated future compute setting, not a used environment. No experimental work is represented as completed.

Pre-document consolidation audit

Audit finding	Handling in this report
The two earlier answers overlap on scope, methods, and gaps.	Scope is summarized once; the full five-gap record appears in Section 3.
Selected prior source checks were reported, but source-opening logs are not reproduced in this report.	Primary-source assertions are described as earlier AI-reported checks, not human-verified findings.
The review is a secondary synthesis; its Table 1 limitations may be reviewer judgments.	Every limitation is attributed to the review unless independently confirmed from the original author text.
Research priority is provisional; novelty has not been proven.	D007 ranking is preserved, with feasibility re-evaluation pending Colab checks.
Official assignment PDF was found through Project file retrieval rather than mounted locally.	Assignment language for Prompt 01/02 was read from page 5; no unverified requirements invented.

2. Prompt 01 - Accountable Research Workspace

2.1 Exact submitted Prompt 01

```
I am a postgraduate student converting my completed literature review into an implementation-based research paper. My topic is Evaluating Tool-Selection Accuracy in Multi-Tool Agentic AI Systems. Act as a critical research mentor and technical pair programmer. Guide me in stages: evidence-based gap identification, literature-to-experiment mapping, feasibility, research questions, verified baseline, coding, experimental evaluation, manuscript and reproducibility. Mark statements as VERIFIED, INFERRED or UNKNOWN, and explicitly request source files or outputs when evidence is missing. Never invent papers, datasets, repository links, code execution, experiments or metrics. Help me create a portable research_decisions.md log containing decisions, reasons, source/evidence and next actions. Start by asking me to attach my existing review and summarize its scope without assuming unseen content.
```

2.2 Scope of the attached literature review

[VERIFIED - review content] The ten-page review [R1] surveys how LLM-based agents decide which external function or API to invoke, separately from whether arguments are correctly formed or actions successfully complete. It organizes the problem into: (i) invocation necessity, (ii) discrimination among candidates, (iii) retrieval from large tool repositories, (iv) argument grounding, and (v) trajectory/order of multiple calls. See review pp. 1, 5-6, Fig. 2 and Section 3.

Methods discussed include Toolformer self-supervised API-use learning, ReAct interleaved reasoning/action, Gorilla retriever-aware training, ToolLLM retrieval plus depth-first exploration, prompting-based tool choice, fine-tuning, and selection-aware reinforcement-learning directions. Benchmarks reviewed include API-Bank, MetaTool, TaskBench,

BFCL, ToolBench, MTU-Bench, TRAJECT-Bench and ToolRet-related retrieval studies. No dataset has yet been downloaded or executed for this project.

The review describes exact/top-1 tool accuracy for unique targets; precision/recall/F1 when multiple tools can be valid; Recall@k and mean reciprocal rank for retrieval; TaskBench Node-F1/Edge-F1 and normalized edit distance for graphs/sequences; and AST-based function-call matching in BFCL. These measure **different stages** and are not directly interchangeable.

2.3 Workspace rules and staged research plan

The working sequence agreed in the first response is: evidence-based gap identification -> literature-to-experiment mapping -> feasibility -> bounded research question -> verifiable baseline -> coding/testing -> experimental evaluation -> manuscript and reproducibility. Only the initial evidence review and gap proposal were undertaken in Prompts 01/02.

Label	Meaning and use
VERIFIED	An inspected artifact establishes the stated fact, with provenance. Example: the review contains a specific claim; this does not verify the original study.
INFERRED	Reasoned interpretation or proposed design to be tested; no measured success implied.
UNKNOWN	Evidence or independent verification missing; requires named source, author text, code, or run logs.

[VERIFIED - conversation] The researcher requested a portable `research_decisions.md`; a local copy exists dated 21 October 2026. It records sources S01-S03, decisions D001-D007, five open questions, next actions and a decision-entry template. Its substance is reproduced in Section 4.

2.4 Prompt 01 substantive decisions and unresolved issues

D001: An implementation-based study must remain anchored in the existing review. D002: Track verified/inferred/unknown evidence and avoid invented datasets, claims or results. D003: Do not finalize a gap before checking the primary literature. D004: Controlled experiments on changing or confusable tools appear feasible **in principle**; no test has established feasibility. The first response requested confirmation of the authoritative review version and possible LaTeX/BibTeX source. Subsequently, the LaTeX ZIP became available. Review version authority remains unconfirmed.

3. Prompt 02 - Evidence-Based Research Gaps

3.1 Exact submitted Prompt 02

```
Analyze my attached literature review Evaluating Tool-Selection Accuracy in Multi-Tool Agentic AI Systems: A Survey of Metrics, Benchmarks, and Open Challenges. Identify central methods, datasets, measured findings, author-reported limitations and unresolved questions. Propose five computationally testable research gaps. For each, cite the specific passage or reference in my review; clearly distinguish a limitation reported in the original paper from your own inference; state the experimental question it suggests and what evidence would test it. Flag any claims needing verification against the original publications. Rank three gaps by research significance, available resources and implementation feasibility. Do not claim novelty without a separate literature search.

Latex is available in the zip file.
```

3.2 Literature-to-evaluation map

Method/benchmark	Methodology	Reported scale/dataset	Measurement or caveat
Toolformer / ReAct	API insertion; interleaved reasoning/action	5 tools for Toolformer; ALFWorld/WebShop and QA settings for ReAct	Task-dependent success; review argues no isolated selection diagnosis
Gorilla / APIBench	Retriever-aware fine-tuning for API calls	1,645 unique APIs; 16,450 instruction pairs (review Table 1)	AST call matching; test-time documentation adaptation
API-Bank	Runnable multi-turn API interactions	73 APIs; annotated dialogues	Calling, retrieval and planning
ToolLLM / ToolBench	Neural retrieval; depth-first exploration	16,464 APIs across 49 categories	Pass rate; win rate; execution simulation caveat
MetaTool / ToolE	Need-to-use and confusable candidate discrimination	4,287 instances; 199 tools reported in review	Selection and no-tool reliability accuracy
TaskBench	Tool graph nodes and dependency edges	Three domains	Node-F1; Edge-F1; normalized edit distance
BFCL	AST matching against reference calls	Thousands of cases; benchmark version varies	Combined function/argument correctness
MTU-Bench / TRAJECT-Bench	Multi-granularity dialogue; trajectory diagnostics	54,798 dialogues / 136 tools for MTU-Bench; trajectory set unspecified	Selection, order, argument and dependency outcomes
ToolRet study	Sparse/dense tool-description retrieval	Review refers to multiple benchmarks; earlier AI response reports 7,600 tasks / 43,000 tools from original abstract	Recall@k; MRR; functional rather than lexical match

Provenance: Scale figures above are verified *as transcribed from review Table 1 and Sections 2-4* [R1, pp. 2-8], not independently validated dataset counts. ToolRet counts were presented as primary-abstract information in the earlier AI response, with source-verification pending review by the student.

3.3 Findings and classification of limitations

Claim in prior work	Classification / verification caution
MLLM-Tool fine-tuning reported 88.19% selection accuracy (review p. 6).	VERIFIED as a review-reported number; UNKNOWN as independently checked result, split and exact metric.
MetaTool exposes sensitivity to overlapping/near-duplicate tools (review pp. 6-8).	Review summary; earlier AI described an official project check, but full original author limitation section not reproduced.
ToolLLM explores tool-use trajectories via DFS; some APIs are simulated (review p. 4).	Original method as summarized in review; simulation criticism is review-table synthesis until author wording checked.
Gorilla retrieval is motivated by changing API documentation (review pp. 7-9).	Earlier AI reported checking its preprint; precise conditions and outcomes still require original experiment details.
TRAJECT-Bench diagnoses tool-versus-argument-versus-order errors (review pp. 4, 6-9).	Earlier AI reported checking original abstract; do not call this diagnostic distinction new.
BFCL AST headline scoring may conceal wrong-tool vs wrong-argument differences (review pp. 6, 8).	Review-author methodological inference; original protocol/diagnostic outputs require code inspection.

Important distinction: A "principal limitation" in the review table is the survey author's analysis unless the cited original paper explicitly self-reports it. The assignment requires that such limitations not be relabeled as original-author admissions.

3.4 Detailed gap matrix - five testable candidates

G1 - Dynamic tool-set churn

Required matrix field	Evidence-based entry
Source passage / reference	[R1] Section 6, p.9: tools added, deprecated, modified; Gorilla [R2].
Author-reported vs inferred	Review-author proposed gap; original Gorilla motivation reported, not author admission of all churn types.
Evidence status	INFERRED gap / original scope UNKNOWN
Unresolved question	How does churn affect selection, and can refreshed retrieval recover performance?
Computational experiment	Paired tool inventories vA/vB: insert, deprecate, replace, update docs; compare fixed-inventory vs refreshed retrieval.
Evidence required	Per-condition top-1 selection, obsolete-tool rate, paired changes, confidence intervals.
Missing verification / limits	Novelty, realistic churn prevalence, labels, original Gorilla protocol unverified.

G2 - Candidate-set confusability

Required matrix field	Evidence-based entry
Source passage / reference	[R1] Sec.3.3 pp.6-7; Sec.5 p.8; MetaTool [R3], TRAJECT-Bench [R4].
Author-reported vs inferred	Difficulty with similar tools is a study finding as reported; cross-benchmark compositional concern is review inference.
Evidence status	Review VERIFIED / independent details incomplete
Unresolved question	At equal set size, do functionally overlapping distractors cause more errors than unrelated ones?
Computational experiment	Matched prompts with identical cardinality; unrelated, similar-name, and overlapping-capability distractors.
Evidence required	Paired accuracy, confusion matrix, stratified errors, multi-valid-label handling.
Missing verification / limits	MetaTool already addresses confusability; new incremental question must be verified.

G3 - Stage-level error attribution

Required matrix field	Evidence-based entry
Source passage / reference	[R1] Sec.5 p.8 and Sec.6 p.9; TaskBench [R5], BFCL [R6], TRAJECT [R4].
Author-reported vs inferred	Aggregation critique is review inference; separate trajectory diagnostics are prior work.
Evidence status	INFERRED extension / novelty UNKNOWN
Unresolved question	What selection errors are hidden by aggregate task success?
Computational experiment	Compare aggregate-only evaluation with separately labeled tool, argument, dependency and execution outcomes; fault injection.
Evidence required	Error decomposition, disagreement of rankings, conditional success rates.
Missing verification / limits	Requires trustworthy traces and adjudication; overlaps established TRAJECT protocols.

G4 - Functional vs lexical retrieval

Required matrix field	Evidence-based entry
Source passage / reference	[R1] Sec.4.2 p.7; Sec.6 p.9; Shi et al. [R7].
Author-reported vs inferred	Standard retriever mismatch is original-study finding as described; designed paraphrase stress test is proposed.
Evidence status	Review VERIFIED; earlier AI source check reported
Unresolved question	Under controlled semantic paraphrases, which retrievers retain correct-tool rank?
Computational experiment	Matched ToolRet subset; query paraphrases; BM25 vs embedding vs capability-based comparator (pending verification).
Evidence required	Recall@1, Recall@k, MRR and rank changes by perturbation.
Missing verification / limits	ToolRet already tests methods; data/license/model access and novel angle unconfirmed.

G5 - Agent-as-tool delegation

Required matrix field	Evidence-based entry
Source passage / reference	[R1] Sec.6 p.9; Tool-RoCo [R8].
Author-reported vs inferred	Limited cooperative use described as Tool-RoCo finding; structured delegation-cost study inferred.
Evidence status	Review VERIFIED / original abstract reported checked
Unresolved question	When is selection of collaborator agents warranted under capability and cost trade-offs?
Computational experiment	Mock specialists with defined competencies; compare direct completion and delegation policy.
Evidence required	Appropriate/over/missed delegation, end-to-end success, latency and cost.
Missing verification / limits	Multi-agent overhead and task-ground-truth design untested; novelty unverified.

3.5 Original top-three ranking (preserved)

Rank	Candidate	Significance	Resources	Feasibility	Original reasoning
1	G1 - Dynamic tool-set changes	5/5	4/5	4/5	Balanced research relevance and manageable controlled simulations; changing APIs are realistic.
2	G2 - Candidate-set confusability	4/5	5/5	5/5	Simplest labeled baseline and interpretable experiment; crowded prior art.
3	G3 - Stage-level error attribution	5/5	4/5	3/5	Useful scientifically but harder traces and labels; trajectory-aware prior art.

[INFERRED - prior AI ranking] These scores are qualitative planning judgments (1-5), not study measurements or novelty scores. In the preceding Prompt 02 response, G1 was provisionally favored, potentially using G2 as a stress condition. That recommendation **did not freeze** an implementation or supersede D003. The user later selected **Google Colab** as the intended compute environment; Colab access, GPU type, packages, quotas and runnable baselines have not been tested.

3.6 Verification and research-quality boundary

Issues explicitly flagged during Prompt 02: reconcile MetaTool model counts (8 in review Table 1 versus an earlier AI-reported repository description of 9); check MLLM-Tool 88.19% metric and evaluation split; inspect original BFCL AST scorer; establish whether candidate-size effects are controlled for semantic similarity; inspect TaskBench/TRAJECT primary limitations; check ToolRet licenses/splits and official code. The ACM Computing Surveys-styled footer of the supplied review is **not** proof of journal publication.

No research novelty is asserted. Dynamic-churn robustness, similarity-controlled selection, error decomposition, tool retrieval and agent delegation are candidate investigative directions only.

4. Consolidated Research Decision Log

Primary artifact: `research_decisions.md` in supplied files (created 21 October 2026). The following records reproduce the substantive decisions and retain their original identifiers. Explanatory wording is tightened for this PDF, not recast as new decisions.

D001 - Implementation-oriented study based on the completed review

Reason	Source/evidence	Verification	Next action
User research objective; stay in subject area	Prompt 01; review [R1]	VERIFIED user intent	Map claims to primary literature.

D002 - Use VERIFIED / INFERRED / UNKNOWN and provenance

Reason	Source/evidence	Verification	Next action
Avoid fabricated claims, outputs and novelty	Prompt 01	VERIFIED workflow rule	Build evidence ledger.

D003 - Keep identified gaps provisional; no frozen question

Reason	Source/evidence	Verification	Next action
Current and primary-source research may overlap	[R1] Sec.6 pp.8-9	VERIFIED review proposals / novelty UNKNOWN	Check original studies and new literature.

D004 - Consider controlled tool-size, similarity or evolving descriptions

Reason	Source/evidence	Verification	Next action
Potentially lower-risk first experiment without tool side effects	[R1] Secs.3,5,6	INFERRED feasibility	Check compute, datasets, scope.

D005 - Use ten-page review as anchor; ZIP as editable source

Reason	Source/evidence	Verification	Next action
Distinguish manuscript content from verified outside evidence	[R1], local ZIP: main.tex/references.bib and 2 figures	VERIFIED artifact presence	Check citations, DOI metadata.

D006 - Track five provisional experimental candidates G1-G5

Reason	Source/evidence	Verification	Next action
Each supplies testable factor and observable outcome	[R1] Secs.3-6; Prompt 02 analysis	INFERRED gaps; novelty UNKNOWN	Check originality and accessible artifacts.

D007 - Rank G1 first, G2 second, G3 third

Reason	Source/evidence	Verification	Next action
Significance, resource availability and feasibility balance	Prompt 02 output and [R1]	INFERRED priority	Confirm Colab feasibility; literature screening; pilot later.

Open questions from the original log

ID	Unresolved item / required input
Q01	Confirm final authoritative review version; ZIP now available but user confirmation outstanding.
Q02	Establish actual novelty from current and original published work.
Q03	Identify, access and reproduce usable benchmark/baseline (not attempted).
Q04	Constrain hardware, budget and timeline: Google Colab stated; runtime details and budget unverified.
Q05	Determine paper venue, grading template and supervisor requirements.

5. AI Prompt and Verification Log

Log item	Prompt 01	Prompt 02
Tool	ChatGPT (assistant identified itself as GPT-6 in earlier response)	ChatGPT
Execution date	21 October 2026 as recorded in research_decisions.md; exact start time not recorded	21 October 2026 according to conversation sequence / log; exact time not recorded
Actual prompt	Verbatim in Section 2.1	Verbatim in Section 3.1
Inputs	User topic, initial literature review PDF in Project, instruction to create portable decisions file	Same review PDF, available LaTeX ZIP, prior workspace/log and existing review discussion
Output	Scope summary, five-part taxonomy, evidence labels, staged plan, D001-D004, research_decisions.md	Comparative methods/datasets, findings and caveats, G1-G5 gap analysis, G1/G2/G3 scores, D005-D007
Verification undertaken	AI inspected review content and file availability; no primary experiment or dataset run	Earlier AI reported selected original-paper/project-page checks; the full verification transcripts are not supplied here
Corrections/refinements	Review claim separated from independently verified publication; later ZIP identified	Explicit caution about review-table limitations and overlapping prior work; unresolved numeric/reference discrepancies
Remaining uncertainty	Authoritative review version, claims vs original source, computational feasibility	Novelty, baseline access, license, measured robustness, exact paper-limitation wording

Human actions not established: There is no evidence the student personally opened and audited original papers, ran code, reproduced baselines or validated an experimental outcome. The earlier AI described selective web checks; this PDF reports those as previous assistant activities only. Future comparison implementation in Colab is a declared resource preference, not an official Prompt 03 execution.

Formatting-stage instruction: The later instruction to consolidate these two prompt responses into a PDF is a document-preparation request, not a third official assignment prompt. Earlier related work consists of writing the ten-page review and creating LaTeX/BibTeX source; further details about AI use in that earlier stage are not substantiated here.

6. Evidence Ledger

Evidence statement	Status	Provenance / action required
Ten-page literature review exists, with taxonomy and five-part evaluation pipeline	VERIFIED (artifact)	[R1] pp. 1, 5-6.
Official assignment specifies Prompt 01 workspace/log and Prompt 02 gap matrix	VERIFIED (assignment text)	[A1] p. 5. Deadline restriction supplied by user.
research_decisions.md exists with D001-D007	VERIFIED (local artifact)	[L1], 21 Oct 2026.
Five candidate gap designs and G1/G2/G3 ranking were produced	VERIFIED (conversation artifact); INFERRED merits	Prompt 02 assistant reply and [L1].
Original studies show various failure modes	VERIFIED as review assertions; primary confirmation partial	[R1] Table 1 and Sections 3-6; check original papers.
MLLM-Tool 88.19% and benchmark scale numbers	VERIFIED as review statements; underlying result UNKNOWN	[R1] pp. 4, 6; inspect splits / actual evaluations.
MetaTool evaluation count 8 versus repository count 9	UNKNOWN discrepancy	Reconcile versions and task subsets.
Specific gap is new; selected approach superior	UNKNOWN	Requires up-to-date novelty search and experiments.
Google Colab available for future use	VERIFIED as user-provided plan; capacity UNKNOWN	User reply; no environment test performed.
No experiment/run outputs completed in these official stages	VERIFIED for available conversation record	No run artifacts / measured project results presented.

7. Conclusion and Next Steps

Prompts 01 and 02 produced an accountable research workspace, a source-dependent summary of tool-selection evaluation, a portable research decision history, five computationally testable hypotheses, and a provisional ranking of G1 (dynamic tools), G2 (confusability), and G3 (error attribution). The review's benchmark and measurement claims were not mistaken for new project results.

Before moving to later assignment stages, independently read prioritized original methods and limitations; confirm official benchmark versions and permitted access; conduct a current novelty search; evaluate Google Colab feasibility; then narrow and approve an executable research question. **No baseline reproduction, code execution or experiment is included in this submission.**

References and Source Inventory

[A1] AI Tools for Research. Post-mid-semester assignment: AI-assisted coding, vibe coding and evidence-based experimentation. Instructor handout, p. 5 (Prompt Studio); uploaded Project copy. Submission deadline and two-prompt restriction are user-supplied, not independently confirmed.

[R1] Mohije, O. A. Evaluating Tool-Selection Accuracy in Multi-Tool Agentic AI Systems: A Survey of Metrics, Benchmarks, and Open Challenges. Supplied 10-page literature review PDF, September 2026; local LaTeX source ZIP. Status: user manuscript, not independently verified as published.

[L1] research_decisions.md. Local research decisions log created 21 October 2026, entries D001-D007, sources and verification warnings.

[R2] Patil, S. G. et al. (2023). Gorilla: Large Language Model Connected with Massive APIs. https://doi.org/10.48550/arXiv.2305.15334

[R3] Huang, Y. et al. (2024). MetaTool Benchmark for Large Language Models: Deciding Whether to Use Tools and Which to Use. https://doi.org/10.48550/arXiv.2310.03128

[R4] He, P. et al. (2025). TRAJECT-Bench: A Trajectory-Aware Benchmark for Evaluating Agentic Tool Use. https://doi.org/10.48550/arXiv.2510.04550

[R5] Shen, Y. et al. (2023/2024). TaskBench: Benchmarking Large Language Models for Task Automation. https://doi.org/10.48550/arXiv.2311.18760

[R6] Patil, S. G. et al. (2025). The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models. https://proceedings.mlr.press/v267/patil25a.html

[R7] Shi, Z. et al. (2025). Retrieval Models Aren't Tool-Savvy: Benchmarking Tool Retrieval for Large Language Models. https://doi.org/10.18653/v1/2025.findings-acl.1258

[R8] Zhang, K. et al. (2025). Tool-RoCo: An Agent-as-Tool Self-Organization Large Language Model Benchmark in Multi-Robot Cooperation. https://doi.org/10.48550/arXiv.2511.21510

[R9] Qin, Y. et al. (2024). ToolLLM: Facilitating Large Language Models to Master 16000+ Real-World APIs. https://doi.org/10.48550/arXiv.2307.16789

[R10] Li, M. et al. (2023). API-Bank: A Comprehensive Benchmark for Tool-Augmented LLMs. https://doi.org/10.18653/v1/2023.emnlp-main.187

[R11] Schick, T. et al. (2023). Toolformer: Language Models Can Teach Themselves to Use Tools. https://doi.org/10.48550/arXiv.2302.04761

[R12] Yao, S. et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. https://doi.org/10.48550/arXiv.2210.03629

[R13] Wang, P. et al. (2024). MTU-Bench: A Multi-Granularity Tool-Use Benchmark for Large Language Models. https://doi.org/10.48550/arXiv.2410.11710

[R14] Wang, C. et al. (2025). MLLM-Tool: A Multimodal Large Language Model for Tool Agent Learning. https://openaccess.thecvf.com/content/WACV2025/html/Wang_MLLM-Tool_A_Multimodal_Large_Language_Model_for_Tool_Agent_Learning_WACV_2025_paper.html

Appendix A - Completeness and Integrity Checklist

Audit criterion	Result
Exact Prompt 01 and 02 reproduced from conversation	PASS
Workspace, review scope and source status recorded	PASS
Exactly five computationally testable candidates	PASS
All five: passage, provenance, question, experiment, evidence, caution	PASS
Original top-three ranking and qualitative scores preserved	PASS
Author limitation versus review inference distinguished	PASS (where source support allows); full original checking outstanding
D001-D007 grounded in existing log	PASS
Dates and validation not fabricated	PASS; earlier source checks labeled as AI reported
Google Colab identified without implying execution	PASS
No later official prompt or experimental output represented as completed	PASS
Novelty claims avoided; reference uncertainties recorded	PASS

Final note: Source citations [R#] identify included bibliography records used in the prior review or discussion. Inclusion does not mean all individual original papers have been re-verified for this submission.